

AASHISH S A

Software Engineer → AI/ML Engineer

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SUMMARY

Software Engineer with ~3 years of experience building enterprise PKI/certificate-lifecycle management and backend systems (Java, Spring Boot, Kubernetes) at AppViewX. Actively transitioning toward AI/ML and GenAI engineering — have shipped a production RAG document-intelligence system and an NLP classifier built from PyTorch primitives, on top of a structured self-study path covering classical ML, deep learning, and Transformers. Co-authored two applied ML/DL research papers in computer vision.

EXPERIENCE

PKI / Certificate Lifecycle Management Engineer — AppViewX

22-01-2024 – Present

- Configured CA integrations (Sectigo, MSCA) for end customers to support 47-day certificate renewal, including auto-renewal and auto-push capabilities.
- Led upgrade projects migrating customers from previous to current product versions, ensuring compatibility, product readiness, and minimal downtime through rigorous testing in live customer environments.
- Delivered 40+ customized feature-request (FR) solutions on 1–2 week cycles per customer, tailoring integrations to each customer's workflow requirements.
- Served as the dedicated engineer for a single enterprise customer for over a year, analyzing and monitoring their full production environment and maintaining 0% downtime across their CLM (Certificate Lifecycle Management) process.
- Resolved 75+ support tickets while providing 24x7 technical support, triaging customer-reported issues with zero prior context and either resolving directly or redirecting to the right team to minimize turnaround time.
- Managed integrations across ITSM tools (ServiceNow and Jira), network management devices (F5, Cloudflare, Imperva, Bluecat, a10, AVI, Thales, SSRS, Cisco) incorporating rate-limiting concepts to ensure stable, reliable automation.
- Collaborated within a 100-member organization, deploying patches directly in customer environments and staying available for new requirements aligned with customer workloads.
- Built enterprise integrations (ServiceNow, F5 BIG-IP, BlueCat DNS) on top of a Certificate Lifecycle Management (CLM) platform automating TLS certificate issuance and rotation.
- Delivered customer-facing automation solutions, working directly with client requirements from scoping through deployment.
- Core stack: Python, Java, Spring Boot, Spring Security, Kubernetes, Docker, REST APIs.

PROJECTS

RAG / Document Intelligence System · Live

- Built a full-lifecycle RAG pipeline turning handwritten notes into cited, grounded answers: Gemini Vision OCR → semantic chunking (300–500 tokens, 50-token overlap) → sentence-transformer embeddings → ChromaDB vector store → top-k retrieval → grounded generation.
- Deployed as a containerized FastAPI service (/query, /health, /metrics) on Railway.
- Stack: Python, FastAPI, Gemini Vision, all-MiniLM-L6-v2, ChromaDB, Docker, Railway

Links: github.com/AashishStark/rag-notes-qa · [Live Application](#)

AG News Text Classifier from Scratch · Live

- Built a 4-class news headline classifier using nn.EmbeddingBag + nn.Linear directly on PyTorch primitives, without legacy torchtext abstractions.
- Implemented a custom tokenizer, vocabulary builder, and Dataset/collate_fn to handle variable-length sequences and offset tensors.
- Stack: PyTorch, nn.EmbeddingBag, custom NLP pipeline

Links: github.com/AashishStark/ag-news-text-classifier · [Colab Link](#)

Notion Python Automation Suite · Live

- Built a modular Python toolkit integrating with the Notion REST API to extract, back up, and batch-update pages across Notes, Reminders, and Travel Planner workspaces. Symmetric Get/Update pattern serializes Notion's nested block tree to local JSON and diff-sync changes back, replacing manual documentation upkeep.
- Stack: Python, Notion API, JSON Schema, CLI

Links: github.com/AashishStark/Notion

Enterprise ERP for Civil Construction · *In Progress*

- Co-building a multi-tenant ERP for civil construction contractors — project milestones, attendance/geofencing, material requisitions, and subcontractor billing — with layered Spring Boot (Controllers/Services/Repositories/DTOs) on PostgreSQL and a React + Vite/Zustand frontend.
- Stack: Spring Boot, React, Vite, PostgreSQL, Docker, Zustand

Links: github.com/satheesh1501/civsaash-backend

SKILLS

Backend & Infrastructure: Java, Spring Boot, Spring Security, REST APIs, Kubernetes, Docker, PostgreSQL, Redis, Linux, Git, CI/CD

Enterprise Integrations: CA CLM Actions, ServiceNow, F5 BIG-IP, BlueCat DNS, PKI / Certificate Automation

AI / ML: Python, NumPy, Pandas, PyTorch, scikit-learn, Transformers & Attention, RAG (ChromaDB, embeddings), FastAPI model serving

PUBLICATIONS

Real-Time Vehicle Pollution Detection and Tracking using Iterative Deep Learning & YOLO

- Co-authored a feedback-based iterative deep learning framework that detects polluting vehicles from CCTV/surveillance footage using an 8-model ensemble (a custom CNN plus 7 pretrained models: Inception-V3, MobileNet-V2/V3, InceptionResNet-V2, VGG16, VGG19, XceptionNet) combined via majority voting.
- Designed a self-training loop that pseudo-labels unlabeled data only when ≥ 5 of 8 models agree above a 0.7 confidence threshold, addressing scarce labeled data; the custom model reached $>90\%$ accuracy vs. 75–85% for the pretrained baselines, and 79.5% in zero-shot evaluation on synthetic rain/fog conditions.
- Integrated YOLOv5 for real-time vehicle detection/localization in video and Gemini AI for license-plate extraction of flagged vehicles, with results logged to Firebase for potential enforcement workflows.

Link: github.com/AashishStark/Vehicle-Pollution-Detection

Plant Disease Recognition with Inception-v3, AlexNet & CNN

- Developed a mobile application that identifies plant diseases from leaf images captured through the device camera or selected from the gallery, using deep learning-based image classification with Inception-v3, AlexNet, and CNN models.
- Implemented disease prediction for multiple plant conditions, including Cotton Healthy Leaf, Tomato Late Blight, and Cedar Apple Rust, and displayed the predicted disease along with relevant treatment/solution information.
- Integrated Google search functionality to help users explore additional information about the detected plant disease and improve accessibility of disease-related knowledge.

Link: github.com/AashishStark/Plant-Disease-with-Inception-v3-Alexnet-CNN

EDUCATION

B.E., Information Technology — Mepco Schlenk Engineering College

2020 – 2024 | CGPA: 8.3